- TDD (04-04-2026)
- Functional programming (08-04-2026)
- Kafka, Spring kafka, Kafka Stream, Kafka connect (11-04-2026)
- Spring Cloud Function, Spring Cloud Stream.(15-04-2026)
- Event, Event Stream, Event Streaming Patterns(18-04-2026)
- Microservice core components.(20-04-2026)
- Security(22-04-2026)

Test Driven Development(TDD)

Scenario:1

- User:I am going to click the button. Should I? Are you sure?
- Dev: ah, yeeees ,click
- User: Oooooh it’s not working . Please check
- Dev: Ok I am checking…..

Scenario:2

Support: Feature x is not working properly

Dev: What? I do this feature and it was workable for last 1 year.

J.Dev Did you make any changes here a few days ago?

J.Dev: Yes, I improved its performance.

Scenario:3

“I said I fixed the spelling error and he said well don't ever do that”

- What?
- Why?
- When?

Testing is an integral part of enterprise software development.

- Catch Bug Early
- Fearless Refactoring
- Documentation That Doesn’t lie
- Better Design
- Faster Development

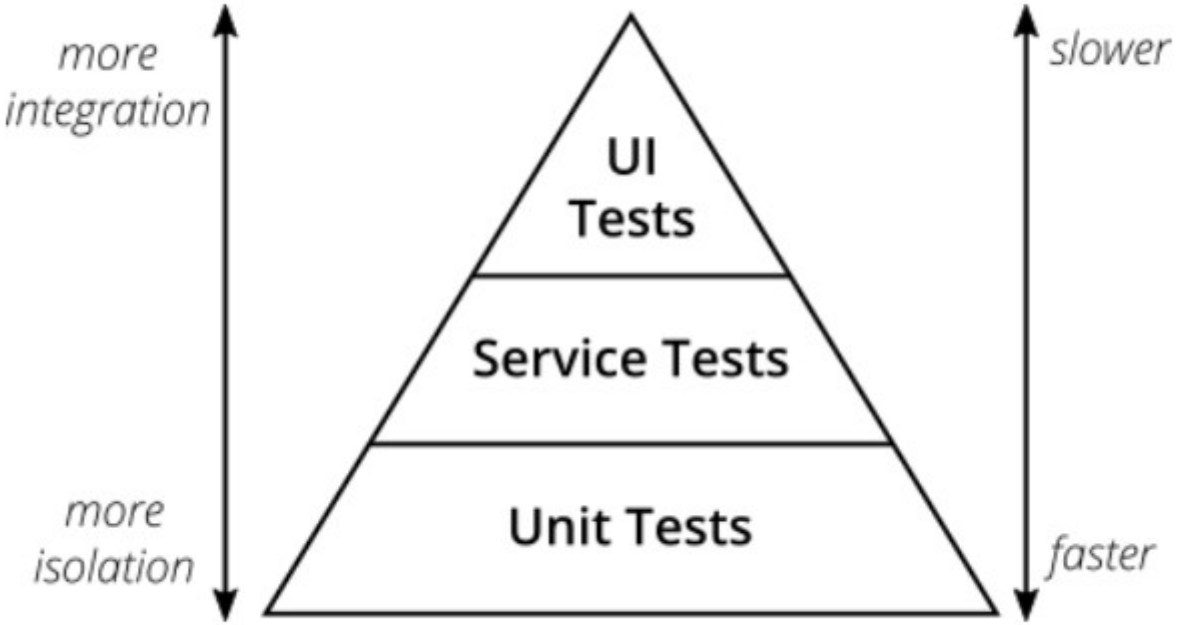
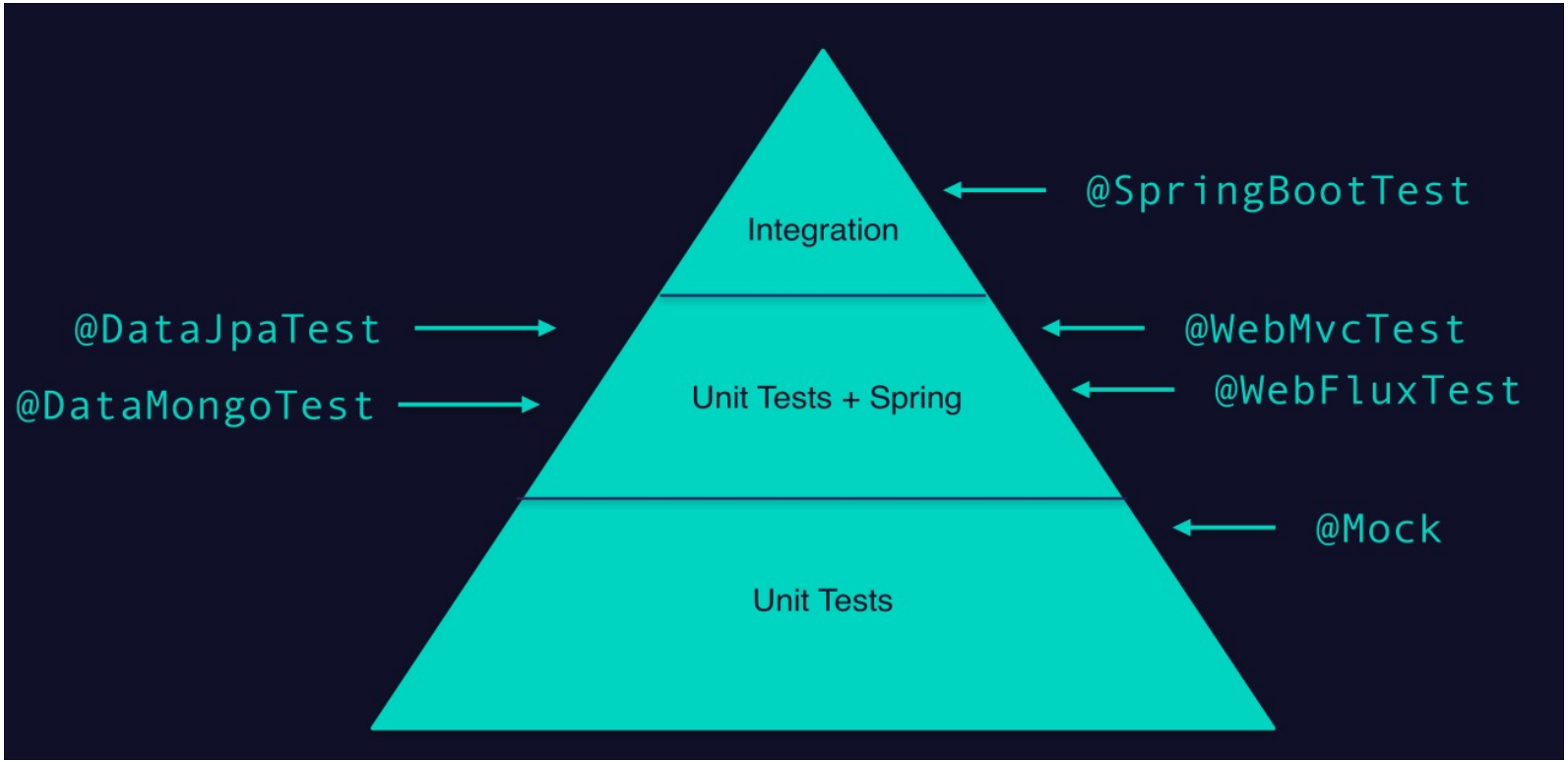


Figure 2: The Test Pyramid


Integration
Unit Tests + Spring
Unit Tests
@SpringBootTest
@DataJpaTest
@DataMongoTest
@WebMvcTest
@WebFluxTest
@Mock

The Three Laws of TDD

- First Law You may not write production code until you have written a failing unit test.
- Second Law You may not write more of a unit test than is sufficient to fail, and not compiling is failing.
- Third Law You may not write more production code than is sufficient to pass the currently failing test.

F.I.R.S.T.

- **Fast** Tests should be fast. They should run quickly.
- **Independent** Tests should not depend on each other.
- **Repeatable** Tests should be repeatable in any environment.
- **Self-Validating** The tests should have a boolean output. Either they pass or fail.

Timely The tests need to be written in a timely fashion.

- Unit tests should be written just before the production code that makes them pass.
- If you write tests after the production code, then you may find the production code to be hard to test.
- You may decide that some production code is too hard to test. You may not design the production code to be testable.

Demo